TAMR-DTI: TARGET-AWARE MODULATION AND MAMBA-BASED REPRESENTATION REFINEMENT FOR DRUG-TARGET INTERACTION PREDICTION

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ABSTRACT

Existing multimodal drug-target interaction (DTI) methods often aggregate molecular conformers as target-independent drug features, so the same ligand geometry is summarized regardless of the paired protein. They also struggle to balance long-range protein representation learning with explicit cross-modal drug-target matching, since generic sequence refinement can weaken the interaction bias required by DTI prediction. To address these limitations, we propose TAMR-DTI, a target-aware multimodal framework that improves a strong BiIntention-based backbone through two complementary innovations. First, a target-aware modulation mechanism uses protein context to reweight drug conformer information and uses fused ligand cues to condition protein token encoding, enabling both modalities to be represented under pair-specific interaction context. Second, a gated Mamba refinement block is applied only to the ordered protein token stream before BiIntention, enhancing protein sequence modeling while preserving explicit drug-target alignment. We evaluate TAMR-DTI on BindingDB, BioSNAP, Human, and C. elegans under repeated-seed train/validation/test splits. The results show improved ranking metrics across all four benchmarks, with the clearest gain on BioSNAP, while the Human benchmark remains a higher-variance setting for fixed-threshold prediction. Ablation and comparative experiments further support the effectiveness of target-aware modulation and protein-side state-space refinement. Overall, TAMR-DTI demonstrates that target-conditioned molecular geometry and protein-only sequence refinement can be jointly integrated to produce a stronger, more interaction-aware framework for DTI prediction.

1 INTRODUCTION

Drug-target interaction (DTI) prediction is a fundamental problem in computational drug discovery because it connects candidate molecules with potential therapeutic targets before expensive biochemical validation. Reliable DTI models can accelerate chemical-genomics screening, target identification, and drug repositioning by narrowing the search space of plausible compound-protein pairs (Keiser et al., 2007; Yamanishi et al., 2008; Liu et al., 2007). The task is challenging because interaction signals depend jointly on molecular chemistry, three-dimensional ligand geometry, protein sequence context, and the pair-specific compatibility between the two modalities. As a result, an effective model should not only encode drugs and targets accurately in isolation, but also preserve the interaction structure that determines whether a given pair is likely to bind.

Existing DTI methods have progressively expanded the representation space used for this task. Early network-inference and similarity-based methods integrated chemical and genomic spaces to infer unknown drug-target links (Yamanishi et al., 2008). Sequence-based deep models such as DeepDTA use neural encoders for SMILES strings and protein sequences, while graph-based models such as GraphDTA encode molecular topology with graph neural networks (Ozturk et al., 2018; Nguyen et al., 2021). Transformer-style and attention-based models further improve compound-protein modeling by learning sequence, substructure, or bilinear interaction patterns (Chen et al., 2020; Huang et al., 2021; Bai et al., 2023; Peng et al., 2025). More recent multimodal frameworks combine pretrained molecular and protein language models, graph encoders, equivariant geometric encoders, and cross-modal fusion modules to exploit complementary drug and target views (Chithrananda et al., 2020; Elnaggar et al., 2022; Satorras et al., 2021; Ji et al., 2026). These developments show that DTI prediction benefits from jointly using sequence semantics, molecular topology, geometric information, and explicit interaction modeling.

Despite this progress, two limitations remain underexplored. First, existing multimodal DTI models usually encode molecular geometry as a drug-side representation before cross-modal fusion (Satorras et al., 2021; Ji et al., 2026); however, the same ligand may expose different relevant conformer cues when paired with different proteins. This weakens the ability of the model to select conformer information under the actual target context. Second, long-range protein sequence modeling and drug-target matching are often entangled in a single fusion design. Although structured state-space models provide efficient long-sequence modeling (Gu et al., 2022; Gu & Dao, 2023), directly applying sequence refinement to mixed drug-protein tokens can blur modality structure, while replacing explicit cross-modal attention may remove the pairwise matching bias that DTI prediction requires. These issues suggest that target-conditioned geometry and protein sequence refinement should be introduced carefully, without disrupting the final interaction module.

To address these limitations, we propose TAMR-DTI, a target-aware multimodal framework built on a strong BiIntention-based DTI backbone. TAMR-DTI introduces bidirectional drug-target modulation: protein context guides conformer-level drug aggregation, and fused ligand information conditions protein token encoding through a lightweight modulation path. In addition, we insert a gated Mamba residual block only on the ordered protein token stream before BiIntention (Gu & Dao, 2023). This design uses state-space modeling where biological token order is meaningful, while preserving BiIntention as the explicit drug-target matching component. In this way, TAMR-DTI strengthens target-aware representation learning and long-range protein refinement without imposing artificial sequential dynamics on heterogeneous drug tokens.

The main contributions of this work are summarized as follows:

- We formulate a target-aware modulation strategy that couples conformer aggregation with protein context and conditions protein token encoding on ligand information.
- We introduce a protein-side Mamba residual refinement module that improves ordered protein representations while retaining explicit BiIntention-based drug-target matching.
- We evaluate TAMR-DTI on BindingDB, BioSNAP, Human, and C. elegans under repeated-seed experimental protocols, showing consistent ranking-metric improvements across all four benchmarks.
- We conduct comparative and ablation studies to verify the effectiveness of target-aware modulation and protein-only state-space refinement within a multimodal DTI framework.

2 Related Work

2.1 DTI Prediction

Computational drug-target interaction (DTI) prediction aims to reduce the experimental search space for identifying potential interactions between small molecules and therapeutic targets. Early chemogenomic studies framed DTI prediction as a link-prediction problem over drug and protein spaces. These methods exploited the observation that chemically similar drugs often bind functionally related proteins, and that target similarity can be inferred from sequence, family, or annotation evidence (Keiser et al., 2007; Yamanishi et al., 2008). Similarity-based inference therefore used ligand similarity, target similarity, and known interaction profiles as the principal evidence for ranking unknown pairs. Kernel learning and regularized matrix-factorization methods further formalized this view by learning latent drug and target factors from sparse interaction matrices while regularizing the solution with neighborhood or profile similarity (Liu et al., 2016; He et al., 2017). Classical machine learning methods, including support vector machines, random forests, nearest-neighbor classifiers, logistic models, and boosting-style predictors, were also widely used when molecular descriptors, protein features, and negative-sampling strategies were available (Cortes & Vapnik, 1995; Breiman, 2001; Guo et al., 2003; Friedman et al., 2000; Liu et al., 2015).

Network-based methods extended pairwise similarity by representing drugs, targets, diseases, side effects, and other biomedical entities in heterogeneous graphs. DTINet learns low-dimensional representations from integrated heterogeneous networks, NeoDTI propagates neighbor information over biomedical relations, and knowledge-graph recommendation models connect DTI prediction to large-scale relational learning (Luo et al., 2017; Wan et al., 2019; Ye et al., 2021). This line of work established that relational context is valuable for DTI prediction, especially when direct interaction observations are sparse. However, similarity, matrix-factorization, and network methods are sensitive to curated similarity definitions, graph coverage, and the quality of observed biomedical links. They also tend to describe drugs and proteins at a relatively coarse level, which limits their ability to model pair-specific molecular substructures, residue-level patterns, and three-dimensional binding geometry.

2.2 Drug Representation

The representation of small-molecule drugs has evolved from hand-crafted chemical features to learned neural representations. Viewed by structural dimensionality, this progression can also be understood as multidimensional small-molecule modeling: one-dimensional SMILES sequences capture linear chemical syntax and substructure semantics, two-dimensional molecular graphs preserve atom-bond topology, and three-dimensional conformer geometry describes spatial arrangements relevant to binding. This 1D/2D/3D view provides the conceptual basis for combining molecular language, topology, and geometry in multimodal DTI models. Traditional cheminformatics pipelines use molecular descriptors to summarize physicochemical properties, topological indices, atom counts, charge-related quantities, and other structure-derived statistics (Todeschini & Consonni, 2009). Molecular fingerprints, including circular or extended-connectivity fingerprints, encode local substructure patterns into fixed-length vectors and have been widely used for ligand similarity search, virtual screening, and classical DTI predictors (Rogers & Hahn, 2010). SMILES strings provide a compact line notation for molecular graphs and enabled sequence-based neural encoders to treat molecules as token sequences (Weininger, 1988). DeepDTA demonstrated that convolutional encoders over SMILES and protein sequences can learn useful representations for binding affinity prediction without relying only on manual features (Ozturk et al., 2018).

Graph-based molecular learning addresses limitations of linear sequence notations by representing atoms as nodes and bonds as edges. Neural graph fingerprints and message-passing neural networks made molecular representation learning differentiable and end-to-end, allowing local atom-bond neighborhoods to be aggregated into task-specific molecular embeddings (Duvenaud et al., 2015; Gilmer et al., 2017). In DTI and affinity prediction, GraphDTA, DGraphDTA, and MGraphDTA use graph neural networks, protein contact maps, or multiscale graph representations to better capture molecular topology and its relation to target binding (Nguyen et al., 2021; Jiang et al., 2020; Yang et al., 2022). More recent molecular encoders incorporate three-dimensional conformers and geometry-aware learning. Continuous-filter networks, equivariant graph neural networks, and related geometric encoders model coordinate-dependent molecular information while respecting rotational and translational structure (Schütt et al., 2017; Satorras et al., 2021). These advances strengthen drug-side representation, but molecular encoders often summarize candidate conformations before the paired protein contributes explicit contextual information.

2.3 Protein Representation

Protein targets are biological macromolecules whose function is determined by amino-acid sequence, local motifs, domains, family membership, structural constraints, and functional sites. In DTI prediction, a protein sequence can indicate residue composition and evolutionary conservation, motifs can mark short functional or binding patterns, domains capture reusable structural and functional units, and protein-family annotations provide higher-level evidence about related targets. Traditional feature engineering therefore used amino-acid composition, dipeptide composition, pseudo-amino-acid composition, composition-transition-distribution descriptors, sequence similarity, and position-specific scoring matrices derived from sequence search (Chou, 2001; Dubchak et al., 1995; Altschul et al., 1997). Domain and functional annotations also became important protein-side features, with Pfam supporting family/domain assignment and Gene Ontology providing structured functional annotation (Finn et al., 2014; Ashburner et al., 2000). These features are biologically interpretable and useful in data-scarce settings, but they depend on external annotation quality and may miss fine-grained residue dependencies that are not captured by fixed descriptors.

Deep learning shifted protein representation from static descriptors to learned sequence encoders. Convolutional models capture local residue motifs and short-range patterns, recurrent models process ordered residue sequences, and Transformer-based architectures model longer-range dependencies through self-attention. DeepConv-DTI emphasized convolution over protein sequences, DeepAffinity combined recurrent and convolutional encoders for compound-protein affinity modeling, and sequence-based DTI models such as TransformerCPI showed that attention can improve compound-protein interaction prediction (Lee et al., 2019; Karimi et al., 2019; Chen et al., 2020). Protein language models further broadened this direction by pretraining on large protein-sequence corpora. ProtTrans and ProtBERT learn contextual protein representations from self-supervised sequence modeling, while ESM shows that large-scale unsupervised learning can encode structural and functional signals from protein sequences (Elnaggar et al., 2022; Rives et al., 2021). These models provide strong general-purpose target embeddings, but their representations are usually learned from protein sequence distributions before a specific drug partner is considered. Recent language-model-based DTI methods further use protein language space as a pair-level modeling substrate. ConPLex learns contrastive drug-protein co-embeddings in protein language space, and DLM-DTI combines dual language models with hint-based learning for DTI prediction (Singh et al., 2023; Lee et al., 2024). These studies reinforce the value of pretrained sequence representations, while also highlighting the need to model the paired drug context rather than relying only on target embeddings learned in isolation.

2.4 Interaction Modeling

After obtaining drug and protein representations, DTI models must determine whether the two modalities are compatible. A common strategy is feature concatenation: a drug embedding and a protein embedding are independently computed, concatenated, and passed to a multilayer predictor. This design is simple and compatible with many encoders, but it leaves the interaction function to infer cross-modal compatibility from compact global vectors. To improve pair-specific modeling, attention and co-attention mechanisms align molecular substructures with protein residues or sequence segments. TransformerCPI uses self-attention for compound-protein interaction prediction, MolTrans learns substructure-level interaction patterns between molecular and protein tokens, and IIFDTI separates independent and interactive features through an attention mechanism (Chen et al., 2020; Huang et al., 2021; Cheng et al., 2022).

Bilinear and bidirectional interaction models further increase the resolution of cross-modal matching. DrugBAN uses a bilinear attention network to capture interpretable drug-target associations and improve transfer across domains, while BINDTI applies bidirectional attention to model mutual interaction intentions between drugs and targets (Bai et al., 2023; Peng et al., 2025). Cross-attention frameworks such as MFCADTI integrate multiple feature sources before prediction and show that interaction modeling benefits from both within-modality representation quality and between-modality alignment (Quan et al., 2025). More recent interaction-centered methods also revisit how pairwise evidence is represented: PSICHIC learns protein-ligand interaction fingerprints from sequence-derived information, and BarlowDTI studies modern one-dimensional DTI prediction with stronger split and representation controls (Koh et al., 2024; Schuh et al., 2025). Overall, attention-based interaction modules have moved DTI prediction beyond late concatenation. A remaining limitation is that many models still perform most pair-specific reasoning only after unimodal drug and protein encoders have already compressed their inputs, which can discard substructure- or residue-level evidence that would be useful for interaction discrimination.

2.5 Multimodal DTI Models

Recent DTI systems increasingly combine language-model features, graph topology, three-dimensional molecular geometry, protein structural information, and cross-modal fusion. Molecular language models such as ChemBERTa provide self-supervised drug sequence features, whereas protein language models such as ProtTrans and ESM provide target embeddings learned from large biological sequence corpora (Chithrananda et al., 2020; Elnaggar et al., 2022; Rives et al., 2021). GraphDTA and its successors incorporate molecular graph topology, and structure-aware methods exploit protein contact maps, multiscale graphs, or geometry-aware molecular encoders to represent structural constraints more explicitly (Nguyen et al., 2021; Jiang et al., 2020; Yang et al., 2022;

Satorras et al., 2021). The rapid progress of protein structure prediction has also made predicted protein structures a more accessible resource for downstream target modeling (Jumper et al., 2021).

Multimodal DTI frameworks seek to combine these complementary views rather than choose a single representation. LDM-DTI integrates pretrained language-model features with graph convolutional and equivariant geometric encoders before dynamic interaction attention (Ji et al., 2026). DrugLAMP emphasizes transferable prediction through pretrained biochemical representations, GEFormerDTA studies transformer graph early fusion for affinity prediction, BiMA-DTI explores a Mamba-Attention hybrid for DTI prediction, and EviDTI introduces evidential learning to quantify predictive uncertainty under multimodal DTI modeling (Luo et al., 2024; Liu et al., 2024; Shui et al., 2025; Zhao et al., 2025). These trends show a clear progression from manually designed similarity and network features toward learned molecular and protein representations, and then toward multimodal interaction modeling. Nevertheless, a general research gap remains: many current pipelines still encode the drug and protein largely independently and perform fusion only at a relatively late stage. As DTI models incorporate sequence semantics, graph topology, conformer geometry, and protein functional context, an important open direction is to make representation learning itself more interaction-aware while preserving the distinct biological and chemical structure of each modality.

3 Methodology

3.1 Problem Definition

We formulate drug-target interaction (DTI) prediction as a supervised binary classification problem. Given a drug molecule D, a target protein T, and a label $y \in \{0, 1\}$ indicating whether the pair interacts, the model learns a scoring function f_θ that estimates the interaction probability:

$$s = f_\theta(D, T), \qquad \hat{y} = p_\theta(y = 1 \mid D, T) = \sigma(s), \tag{1}$$

where s is the prediction logit, $\sigma(\cdot)$ is the sigmoid function, and θ denotes all trainable parameters. For a mini-batch $\mathcal{B} = \{(D_i, T_i, y_i)\}_{i=1}^{B}$, the learning objective is to assign larger logits to interacting pairs and smaller logits to non-interacting pairs.

Each drug is described by three complementary views. The first view is a SMILES-level molecular language representation extracted from ChemBERTa (Chithrananda et al., 2020). The second view is a two-dimensional molecular graph, which preserves atom-level topology and chemical bonds as in graph-based DTI models (Nguyen et al., 2021). The third view is a set of three-dimensional conformers encoded with an equivariant geometric network (Satorras et al., 2021). Each target protein is represented by residue-level ProtBERT features from the ProtTrans family (Elnaggar et al., 2022). The model must encode each modality, align drug geometry with the paired target, and perform explicit drug-protein matching.

Throughout this section, $d = 128$ denotes the shared hidden dimension. The drug graph contains at most $N_g = 290$ atom tokens, each conformer contains $N_c = 64$ resampled atom tokens, and the main model uses $K = 8$ conformers. The drug token length after topology-geometry concatenation is $L_d = N_g + N_c = 354$. The protein sequence length is denoted by L_p, with a binary mask $\mathbf{m} \in \{0, 1\}^{L_p}$ used to identify valid residues.

3.2 Model Overview

TAMR-DTI is built on a BiIntention-based DTI backbone (Peng et al., 2025; Ji et al., 2026). Pretrained molecular and protein features provide semantic priors, graph convolution captures molecular topology, equivariant geometry models conformer structure, and BiIntention performs cross-modal matching. TAMR-DTI adds two pair-specific refinements. Protein context reweights drug conformers, making geometric aggregation target conditioned. A gated Mamba residual block is then applied only to the ordered protein token stream before BiIntention (Gu et al., 2022; Gu & Dao, 2023).

Figure 1 shows the TAMR-DTI architecture, including the three drug streams, the target-aware conformer weighting path, ligand-conditioned protein encoding, protein-side Mamba refinement, BiIntention-based cross-modal matching, and final interaction prediction.

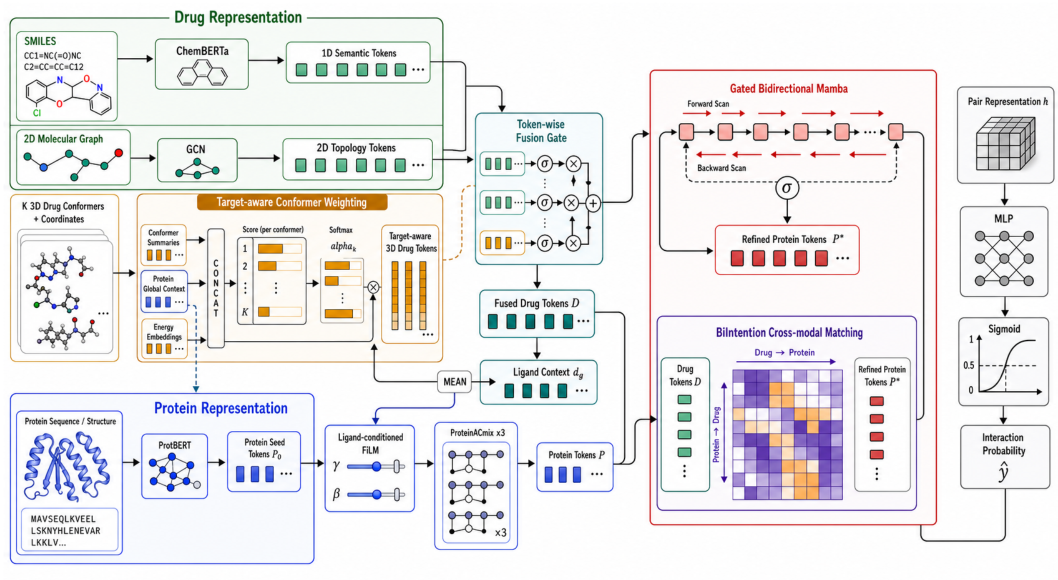

Figure 1: TAMR-DTI architecture. The model integrates SMILES semantics, molecular topology, and target-aware conformer geometry on the drug side; uses ligand-conditioned FiLM and protein-side Mamba refinement on the protein side; and performs final drug-target matching through BiIntention before MLP-based interaction prediction.

The computation is

$$
\begin{aligned}
\mathbf{D} &= \mathrm{DrugEnc}\big(\mathbf{X}^{1D}, \mathcal{G}_D, \{\mathbf{F}_k, \mathbf{R}_k, \epsilon_k, q_k\}_{k=1}^{K}, \mathbf{P}^0, \mathbf{m}\big)\,, \\
\mathbf{d}_g &= \mathrm{MeanPool}(\mathbf{D}), \\
\mathbf{P} &= \mathrm{ProtEnc}\big(\mathbf{P}^0, \mathbf{m}, \mathbf{d}_g\big)\,, \\
\tilde{\mathbf{P}} &= \mathrm{ProtMamba}(\mathbf{P}, \mathbf{m}), \\
\mathbf{h} &= \mathrm{BiIntention}(\mathbf{D}, \tilde{\mathbf{P}}), \\
s &= \mathrm{MLP}(\mathbf{h}), \qquad \hat{y} = \sigma(s).
\end{aligned} \tag{2}
$$

Here $\mathbf{X}^{1D} \in \mathbb{R}^{L_d \times d}$ denotes the ChemBERTa drug feature sequence, $\mathcal{G}_D$ is the molecular graph, $(\mathbf{F}_k, \mathbf{R}_k)$ are the atom features and coordinates of conformer k, ϵ_k is its conformer energy, and $q_k \in \{0, 1\}$ marks whether the conformer is valid. $\mathbf{P}^0$ is the projected ProtBERT protein feature, $\mathbf{D} \in \mathbb{R}^{L_d \times d}$ is the fused drug token sequence, $\mathbf{P} \in \mathbb{R}^{L_p \times d}$ is the ligand-conditioned protein token sequence before protein-side Mamba refinement, $\tilde{\mathbf{P}}$ is the refined protein token sequence, $\mathbf{d}_g$ is the global ligand context used for ligand-conditioned protein encoding, and $\mathbf{h} \in \mathbb{R}^{2d}$ is the fused pair representation. The projected protein sequence first guides conformer aggregation, and the resulting drug representation then provides ligand context for protein encoding before final cross-modal matching.

3.3 Drug Encoder

The drug encoder integrates molecular language, graph topology, and target-aware 3D geometry. SMILES features encode chemical semantics learned from molecular corpora, graph features preserve atom-neighbor connectivity, and conformer features encode spatial arrangement. TAMR-DTI computes topology and geometry tokens, then fuses them with the ChemBERTa token sequence through a learnable gate.

Molecular topology. Let $\mathcal{G}_D = (\mathcal{V}, \mathcal{E})$ be the padded molecular graph of drug D, where each atom node has an input feature $\mathbf{a}_i \in \mathbb{R}^{75}$. The implementation first projects atom features to the hidden space:

$$\mathbf{h}_i^{g,0} = \mathbf{W}_g \mathbf{a}_i, \qquad \mathbf{H}^{g,0} \in \mathbb{R}^{N_g \times d}. \tag{3}$$

When graph padding is enabled, the last input channel is used as the virtual padding marker, and its projection weight is fixed to zero so that padded nodes do not introduce artificial atom semantics. A multi-layer graph convolutional network then aggregates local chemical topology:

$$\mathbf{H}^g = \mathrm{GCN}(\mathcal{G}_D, \mathbf{H}^{g,0}) \in \mathbb{R}^{N_g \times d}. \tag{4}$$

The resulting $\mathbf{H}^g$ provides atom-level topology tokens.

Conformer geometry. For conformer k, let $\mathbf{F}_k \in \mathbb{R}^{N_c \times d}$ denote atom features and $\mathbf{R}_k \in \mathbb{R}^{N_c \times 3}$ denote Cartesian coordinates. TAMR-DTI uses an EGNN layer to update atom features while respecting Euclidean symmetry:

$$(\mathbf{Z}_k, \mathbf{R}'_k) = \mathrm{EGNN}(\mathbf{F}_k, \mathbf{R}_k), \qquad \mathbf{Z}_k \in \mathbb{R}^{N_c \times d}. \tag{5}$$

The conformer-level summary is obtained by average pooling over atom tokens:

$$\mathbf{c}_k = \frac{1}{N_c} \sum_{i=1}^{N_c} \mathbf{z}_{k,i}. \tag{6}$$

This summary is used only to score conformer relevance; the final 3D drug representation remains atom-resolved.

The conformer slots are kept in the order produced by preprocessing. RDKit ETKDG is used to generate candidate conformers, UFF optimization is applied when force-field parameters are available, and the resulting conformers are sorted by their computed UFF energy before padding or truncation. Therefore, the first slot corresponds to the lowest computed-energy conformer among the generated candidates, slots two to four provide lower-energy alternative conformations, and later slots contain higher-energy generated alternatives. The energy value is used as an auxiliary conformer-ranking feature rather than as a calibrated cross-molecule thermodynamic quantity. The conformer-count comparison follows this order: the $n = 1$ setting tests a fixed lowest-computed-energy geometry, whereas larger n values expose progressively broader geometric alternatives to the target-aware selector.

Target-aware conformers. Conformer aggregation depends on the paired protein. The same ligand can expose different binding-relevant geometric cues for different targets, so a fixed conformer average may dilute target-specific 3D signal. ProtBERT tokens are first normalized and projected:

$$\mathbf{P}^0 = \mathbf{W}_p \ \mathrm{LayerNorm}(\mathbf{P}^{bert}), \qquad \mathbf{P}^{bert} \in \mathbb{R}^{L_p \times 1024}, \quad \mathbf{P}^0 \in \mathbb{R}^{L_p \times d}. \tag{7}$$

The masked protein context vector is then

$$\mathbf{p}_g = \frac{\sum_{j=1}^{L_p} m_j \mathbf{p}_j^0}{\max(1, \sum_{j=1}^{L_p} m_j)}. \tag{8}$$

For each conformer, its energy ϵ_k is projected to the hidden space, $\mathbf{u}_k = \mathbf{W}_e \epsilon_k$. The conformer logit combines the conformer summary, the protein context, and the energy embedding:

$$a_k = \mathbf{w}_a^\top \, \mathrm{SiLU}\left(\mathbf{W}_a[\mathbf{c}_k; \mathbf{p}_g; \mathbf{u}_k] + \mathbf{b}_a\right) + b_s. \tag{9}$$

Invalid conformers are masked by assigning their logits a large negative value. For samples with at least one valid conformer, the conformer weights are

$$\alpha_k = \frac{\exp(a_k)\, q_k}{\sum_{\ell=1}^{K} \exp(a_\ell)\, q_\ell}, \qquad \sum_{k=1}^{K} \alpha_k = 1. \tag{10}$$

In the rare case that preprocessing yields an empty conformer-validity mask, the implementation activates the first conformer slot as a fallback before applying the softmax, preventing a zero denominator in Eq. 10. The target-aware atom-level 3D representation and its global context are

$$\mathbf{Z}^{3D} = \sum_{k=1}^{K} \alpha_k \mathbf{Z}_k, \qquad \mathbf{c}^{3D} = \sum_{k=1}^{K} \alpha_k \mathbf{c}_k. \tag{11}$$

Thus, conformer weights are conditioned on the paired protein rather than fixed for the ligand alone.

Drug fusion. The graph topology tokens and target-aware 3D tokens are concatenated:

$$\mathbf{X}^{geo} = [\mathbf{H}^g; \mathbf{Z}^{3D}] \in \mathbb{R}^{(N_g+N_c)\times d}. \tag{12}$$

This sequence has the same length as the cached ChemBERTa feature sequence $\mathbf{X}^{1D} \in \mathbb{R}^{L_d\times d}$. In preprocessing, the ChemBERTa hidden states are mapped to the same fixed drug-token length $L_d = 354$ used by the structural branch. Therefore, the fusion gate should be understood as a fixed latent-slot operation over two equal-length drug representations after preprocessing. The slot index does not imply a one-to-one chemical alignment between a SMILES token and a graph atom. Both branches are normalized and combined by a slot-wise, channel-wise gate:

$$\begin{aligned}
\tilde{\mathbf{X}}^{1D} &= \mathrm{LayerNorm}(\mathbf{X}^{1D}),\\
\tilde{\mathbf{X}}^{geo} &= \mathrm{LayerNorm}(\mathbf{X}^{geo}),\\
\mathbf{G}^{D} &= \sigma\Big(\mathbf{W}_f[\tilde{\mathbf{X}}^{1D}; \tilde{\mathbf{X}}^{geo}] + \mathbf{b}_f\Big),\\
\mathbf{D} &= \mathbf{G}^{D} \odot \tilde{\mathbf{X}}^{1D} + (1-\mathbf{G}^{D}) \odot \tilde{\mathbf{X}}^{geo}.
\end{aligned} \tag{13}$$

The gate bias is initialized to -2.0, so the initial value $\sigma(-2.0) \approx 0.119$ assigns a small but nonzero weight to the ChemBERTa branch and a larger initial weight to the topology-geometry branch. During training, the gate learns the token-wise balance between pretrained molecular semantics and structural information. The global ligand context used by the protein module is

$$\mathbf{d}_g = \frac{1}{L_d}\sum_{i=1}^{L_d} \mathbf{d}_i. \tag{14}$$

3.4 Protein Encoder

The protein encoder maps ProtBERT residue features into ligand-conditioned protein tokens. ProtBERT provides residue-level sequence priors, and the FiLM path lets the paired ligand modulate these tokens before cross-modal fusion.

The input protein sequence is first mapped from the ProtBERT dimension to the shared hidden dimension using Eq. 7. The projected tokens $\mathbf{P}^0$ are passed through three ProteinACmix layers. Each layer combines a feed-forward half-step, ligand-conditioned feature modulation, relative self-attention, dynamic convolution, and a second feed-forward half-step. For layer $\ell = 1, \ldots, 3$, let $\mathbf{P}^{\ell-1} \in \mathbb{R}^{L_p\times d}$ be the input tokens. The first feed-forward residual update is

$$\mathbf{U}^{\ell} = \frac{1}{2}\,\mathrm{FFN}_1^{\ell}(\mathbf{P}^{\ell-1}) + \frac{1}{2}\mathbf{P}^{\ell-1}. \tag{15}$$

If ligand conditioning is enabled, the global drug context $\mathbf{d}_g$ is converted into feature-wise scale and shift parameters $\boldsymbol{\gamma}^{\ell}, \boldsymbol{\beta}^{\ell} \in \mathbb{R}^d$:

$$\begin{aligned}
[\boldsymbol{\gamma}^{\ell}, \boldsymbol{\beta}^{\ell}] &= \mathrm{MLP}_{film}^{\ell}(\mathbf{d}_g),\\
\bar{\boldsymbol{\gamma}}^{\ell} &= \rho \tanh(\boldsymbol{\gamma}^{\ell}), \qquad\qquad \rho = 0.05.\\
\bar{\boldsymbol{\beta}}^{\ell} &= \rho \boldsymbol{\beta}^{\ell},
\end{aligned} \tag{16}$$

The protein tokens are then modulated as

$$\tilde{\mathbf{U}}^{\ell} = \mathbf{U}^{\ell} \odot (1 + \bar{\boldsymbol{\gamma}}^{\ell}) + \bar{\boldsymbol{\beta}}^{\ell}. \tag{17}$$

where the scale and shift vectors are broadcast over the residue-token axis. The fixed scale ρ keeps ligand modulation small at initialization.

The modulated tokens are further processed by the CNN-Transformer block:

$$\begin{aligned}
\mathbf{A}^{\ell} &= \mathrm{MHSA}_{rel}^{\ell}(\tilde{\mathbf{U}}^{\ell}, \mathbf{m}) + \tilde{\mathbf{U}}^{\ell},\\
\mathbf{C}^{\ell} &= \mathrm{Conv}^{\ell}(\mathbf{A}^{\ell}) + \mathbf{A}^{\ell},\\
\mathbf{P}^{\ell} &= \frac{1}{2}\,\mathrm{FFN}_2^{\ell}(\mathbf{C}^{\ell}) + \frac{1}{2}\mathbf{C}^{\ell}.
\end{aligned} \tag{18}$$

The relative multi-head self-attention captures long-range residue dependencies under the protein mask, and the convolutional branch emphasizes local sequence motifs that may be relevant to interaction prediction. After three layers, the protein module outputs

$$\mathbf{P} = \mathbf{P}^3 \in \mathbb{R}^{L_p\times d}. \tag{19}$$

3.5 ALIGNMENT AND MASKS

The encoders produce cached features with different dimensions and scales. Before fusion, ChemBERTa features, atom descriptors, conformer features, and ProtBERT residue features are mapped into a shared hidden space.

TAMR-DTI uses three alignment operations. First, all active streams are mapped to the same hidden dimension $d = 128$:

$$\psi_{1D} : \mathbb{R}^{128} \to \mathbb{R}^{128}, \qquad \psi_g : \mathbb{R}^{75} \to \mathbb{R}^{128}, \qquad \psi_p : \mathbb{R}^{1024} \to \mathbb{R}^{128}. \tag{20}$$

Here ψ_{1D} is implemented as layer normalization over the cached ChemBERTa sequence, ψ_g is the graph input projection in Eq. 3, and ψ_p is the ProtBERT projection in Eq. 7. Second, masks are used at the stages where the implementation explicitly supports them. The protein mask $\mathbf{m}$ is used in masked mean pooling, relative self-attention inside the protein encoder, and protein-side Mamba refinement. The final BiIntention block operates on the fixed-length protein token sequence produced by the protein encoder. The conformer validity mask $\mathbf{q} = (q_1, \ldots, q_K)$ prevents invalid conformers from receiving probability mass in Eq. 10. Third, pair-level context vectors are constructed by pooling aligned tokens:

$$\mathbf{d}_g = \mathrm{mean}(\mathbf{D}), \qquad \mathbf{p}_g = \mathrm{MaskedMean}(\mathbf{P}^0, \mathbf{m}). \tag{21}$$

The aligned interface is $\{\mathbf{D}, \mathbf{P}, \mathbf{m}, \boldsymbol{\alpha}\}$, where $\boldsymbol{\alpha}$ records the target-aware conformer weights.

3.6 FUSION

Fusion combines protein-side sequence refinement with BiIntention-based drug-target matching. The main configuration uses `protein_mamba_biintention`: Mamba is applied only to the protein token stream, and BiIntention remains the pairwise interaction module. Protein residues form an ordered biological sequence, whereas drug tokens concatenate SMILES semantics, graph topology, and conformer geometry.

Protein Mamba. Given protein tokens $\mathbf{P}$ and mask $\mathbf{m}$, padded positions are suppressed before entering the Mamba residual block:

$$\mathbf{P}_j^{mask} = m_j \mathbf{p}_j. \tag{22}$$

The Mamba residual block normalizes the sequence, applies a forward selective state-space mixer, and applies a backward mixer to the reversed sequence:

$$\begin{aligned} \mathbf{H} &= \mathrm{LayerNorm}(\mathbf{P}^{mask}), \\ \mathbf{M}_f &= \mathrm{Mamba}_f(\mathbf{H}), \\ \mathbf{M}_b &= \mathrm{Reverse}(\mathrm{Mamba}_b(\mathrm{Reverse}(\mathbf{H}))), \\ \mathbf{M} &= \mathbf{W}_m[\mathbf{M}_f; \mathbf{M}_b]. \end{aligned} \tag{23}$$

The block then uses residual and feed-forward updates:

$$\begin{aligned} \mathbf{R} &= \mathbf{P}^{mask} + \mathrm{Dropout}(\mathbf{M}), \\ \mathbf{P}^m &= \mathbf{R} + \mathrm{FFN}_m(\mathbf{R}). \end{aligned} \tag{24}$$

A scalar gate controls how much of the refined protein representation is injected:

$$\tau = \sigma(\eta), \qquad \tilde{\mathbf{P}} = \mathbf{P} + \tau(\mathbf{P}^m - \mathbf{P}). \tag{25}$$

The gate logit is initialized as $\eta = -2.0$, giving $\tau \approx 0.119$. The mask is applied inside the Mamba residual block; the subsequent BiIntention module receives the resulting fixed-length protein token sequence.

BiIntention. After protein-side refinement, TAMR-DTI uses BiIntention to model explicit drug-protein compatibility. Self-attention is first applied independently to each modality:

$$\mathbf{D}^s = \mathrm{SA}_D(\mathbf{D}), \qquad \mathbf{P}^s = \mathrm{SA}_P(\tilde{\mathbf{P}}), \tag{26}$$

where each self-attention module includes a residual connection. The intention operator maps a source sequence $\mathbf{X} \in \mathbb{R}^{n_x \times d}$ and a query sequence $\mathbf{Q} \in \mathbb{R}^{n_q \times d}$ to a compact cross-modal intention representation. We set $\mathbf{D}^{(0)} = \mathbf{D}^s$ and $\mathbf{P}^{(0)} = \mathbf{P}^s$ for the following bidirectional update. For attention head r with head dimension $d_h = d/H$,

$$\begin{aligned}
\mathbf{Q}_r &= \mathbf{Q}\mathbf{W}_r^Q, \\
\mathbf{K}_r &= \mathbf{X}\mathbf{W}_r^K, \\
\mathbf{V}_r &= \mathbf{X}\mathbf{W}_r^V, \\
\mathbf{A}_r &= \omega_r \ \mathrm{softmax}_{d_h}\left((\lambda\mathbf{I} + \mathbf{K}_r^\top\mathbf{K}_r)^\dagger\mathbf{K}_r^\top\mathbf{V}_r\right), \\
\mathbf{O}_r &= \mathbf{Q}_r\mathbf{A}_r,
\end{aligned} \tag{27}$$

where λ is a shared learnable ridge parameter, ω_r is a learnable head weight, $(\cdot)^\dagger$ denotes the Moore-Penrose pseudoinverse, and $\mathrm{softmax}_{d_h}$ follows the implementation's head-feature axis. Concatenating heads and applying an output projection gives $\mathrm{Int}(\mathbf{X}, \mathbf{Q}) \in \mathbb{R}^{n_q \times d}$. The intention block then computes

$$\Phi(\mathbf{X}, \mathbf{Q}) = \beta\, \mathbf{Q}^\top \mathrm{softmax}\left(\mathrm{Int}(\mathrm{LayerNorm}(\mathbf{X}), \mathbf{Q})\right), \tag{28}$$

with learnable scalar β, yielding a tensor in $\mathbb{R}^{d \times d}$. For the ℓ-th BiIntention layer,

$$\begin{aligned}
\mathbf{V}_p^{(\ell)} &= \Phi(\mathbf{D}^{(\ell-1)}, \mathbf{P}^{(\ell-1)}), \\
\mathbf{V}_d^{(\ell)} &= \Phi(\mathbf{P}^{(\ell-1)}, \mathbf{D}^{(\ell-1)}), \\
\mathbf{D}^{(\ell)} &= \mathbf{V}_d^{(\ell)}, \qquad \mathbf{P}^{(\ell)} = \mathbf{V}_p^{(\ell)}.
\end{aligned} \tag{29}$$

The main model uses one BiIntention layer and eight attention heads. The final BiIntention tensors are not used as replacement unimodal encodings; instead, they are compact bidirectional interaction representations from which the drug-side and protein-side pair features are pooled. The implementation applies max pooling over the final fixed-length BiIntention tensors:

$$\mathbf{h}_d = \mathrm{MaxPool}(\mathbf{D}^{(L)}), \qquad \mathbf{h}_p = \mathrm{MaxPool}(\mathbf{P}^{(L)}), \qquad \mathbf{h} = [\mathbf{h}_d; \mathbf{h}_p] \in \mathbb{R}^{2d}. \tag{30}$$

3.7 Prediction Head

The prediction head maps the fused pair representation $\mathbf{h} \in \mathbb{R}^{256}$ to a scalar interaction logit. It is implemented as a four-layer MLP with layer normalization after the first three hidden transformations:

$$\begin{aligned}
\mathbf{z}_1 &= \mathrm{LayerNorm}\left(\mathrm{ReLU}(\mathbf{W}_1\mathbf{h} + \mathbf{b}_1)\right), \quad \mathbf{z}_1 \in \mathbb{R}^{256}, \\
\mathbf{z}_2 &= \mathrm{LayerNorm}\left(\mathrm{ReLU}(\mathbf{W}_2\mathbf{z}_1 + \mathbf{b}_2)\right), \quad \mathbf{z}_2 \in \mathbb{R}^{256}, \\
\mathbf{z}_3 &= \mathrm{LayerNorm}\left(\mathrm{ReLU}(\mathbf{W}_3\mathbf{z}_2 + \mathbf{b}_3)\right), \quad \mathbf{z}_3 \in \mathbb{R}^{128}, \\
s &= \mathbf{W}_4\mathbf{z}_3 + \mathbf{b}_4.
\end{aligned} \tag{31}$$

The predicted probability is obtained by applying the sigmoid function as in Eq. 1.

3.8 Training Objective

The active training objective is weighted binary cross-entropy with logits. For sample i, let s_i be the model logit and $\hat{y}_i = \sigma(s_i)$. The mini-batch loss is

$$\mathcal{L}_{BCE} = -\frac{1}{B}\sum_{i=1}^{B}\left[w_+ y_i \log \hat{y}_i + (1 - y_i)\log(1 - \hat{y}_i)\right], \tag{32}$$

where w_+ is the positive-class weight. When automatic positive weighting is enabled, the weight is computed from the training split as

$$w_+ = \frac{N_-}{N_+}, \tag{33}$$

where N_+ and N_- are the numbers of positive and negative training samples.

All reported main experiments optimize Eq. 32 directly. Checkpoint selection, optimizer settings, learning-rate scheduling, early stopping, and threshold conventions are reported in Section 4.1.3. In particular, the main results select checkpoints by validation AUROC; fixed-threshold metrics are reported as Acc@0.5 and F1@0.5, while validation-selected operating points, when used, are reported separately as Acc(val-thr) and F1(val-thr).

4 Experiments

4.1 Experimental Setup

This section describes the datasets, evaluation metrics, and implementation settings used for the main TAMR-DTI experiments. All main results use the full eight-conformer TAMR-DTI model and five fixed random seeds, $\{42, 43, 44, 45, 47\}$. For every run, the best checkpoint is selected by validation AUROC. When a threshold-dependent result uses a tuned operating point, the classification threshold is selected only on the validation split and is then applied unchanged to the held-out test split; such results are reported as Acc(val-thr) and F1(val-thr). Fixed-threshold results, when reported, are denoted Acc@0.5 and F1@0.5.

4.1.1 Datasets and Data Distribution

We evaluate TAMR-DTI on four benchmark drug-target interaction (DTI) datasets: BindingDB, BioSNAP, Human, and C. elegans. Each sample consists of a drug SMILES string, a protein sequence, and a binary interaction label. Table 1 reports the aggregate statistics after merging the processed training, validation, and test split files used in this study; the corresponding split-level statistics are moved to Appendix Table 9. These processed-split statistics are used throughout the experiments, because they reflect the exact drug-target pairs available to the model after preprocessing.

BindingDB is a public database of experimentally measured protein-ligand binding affinities and drug-target interactions (Liu et al., 2007). In our processed splits, it is the largest benchmark, with 49,199 labeled interactions, 14,643 drugs, and 2,623 proteins. Its positive rate is 42.02%, giving a moderate negative-label skew. We use BindingDB to test whether TAMR-DTI can scale to broad chemical and target coverage while remaining robust under non-balanced label distributions.

BioSNAP comes from the Stanford Biomedical Network Dataset Collection and provides curated biomedical network data, including drug-target associations (Zitnik et al., 2018). The processed BioSNAP benchmark is medium-sized, with 27,464 interactions, and is nearly class-balanced, with a 50.36% positive rate. Compared with BindingDB, it offers a different curation source and a more balanced decision setting. We therefore use BioSNAP to evaluate generalization under a network-derived DTI distribution and to support the component analyses in later sections.

The Human benchmark follows the compound-protein interaction dataset introduced with high-confidence negative sample construction (Liu et al., 2015). It is the smallest benchmark in our study, with 5,997 interactions and a 43.91% positive rate. Its compact benchmark size and stronger negative skew make it useful for testing model stability when fewer training examples are available and thresholded metrics are more sensitive to class distribution.

The C. elegans benchmark uses the corresponding Caenorhabditis elegans compound-protein interaction data from the same dataset family (Liu et al., 2015). The processed split contains 7,786 interactions and is exactly balanced in aggregate. Because it represents a non-human model organism, it complements the Human benchmark and tests whether the learned drug and protein representations transfer across organism-specific interaction sources rather than only across larger mixed-source datasets.

Table 1: Dataset statistics used in the main experiments. Counts are computed from the processed training, validation, and test split files used in this study.

Dataset	Drugs	Proteins	Interactions	Positive	Negative	Positive rate
BindingDB	14,643	2,623	49,199	20,674	28,525	42.02%
BioSNAP	4,505	2,181	27,464	13,830	13,634	50.36%
Human	2,726	2,001	5,997	2,633	3,364	43.91%
C. elegans	1,767	1,876	7,786	3,893	3,893	50.00%

4.1.2 Evaluation Metrics

We report two threshold-independent ranking metrics, AUROC and AUPRC, and two threshold-dependent classification metrics, Acc@0.5 and F1@0.5. For a binary DTI prediction task, true positives (TP) are interacting drug-target pairs correctly predicted as positive, true negatives (TN) are non-interacting pairs correctly predicted as negative, false positives (FP) are non-interacting pairs predicted as positive, and false negatives (FN) are interacting pairs predicted as negative.

AUROC is the area under the receiver operating characteristic (ROC) curve. For a decision threshold τ, the true positive rate and false positive rate are

$$\mathrm{TPR}(\tau) = \frac{\mathrm{TP}(\tau)}{\mathrm{TP}(\tau) + \mathrm{FN}(\tau)}, \qquad \mathrm{FPR}(\tau) = \frac{\mathrm{FP}(\tau)}{\mathrm{FP}(\tau) + \mathrm{TN}(\tau)}. \tag{34}$$

The ROC curve varies τ over the prediction-score range, so AUROC measures positive-negative ranking quality independent of any single threshold. AUROC therefore is not reported with an @0.5 suffix.

AUPRC is the area under the precision-recall curve. For threshold τ,

$$\mathrm{Precision}(\tau) = \frac{\mathrm{TP}(\tau)}{\mathrm{TP}(\tau) + \mathrm{FP}(\tau)}, \qquad \mathrm{Recall}(\tau) = \frac{\mathrm{TP}(\tau)}{\mathrm{TP}(\tau) + \mathrm{FN}(\tau)}. \tag{35}$$

AUPRC emphasizes retrieval quality for the positive interaction class and is useful when positive-label rates differ across datasets. Like AUROC, AUPRC is computed across thresholds and is not an @0.5 metric.

For fixed-threshold classification metrics, we convert a predicted interaction probability p_i into a binary prediction using $\hat{y}_i^{0.5} = \mathbb{I}(p_i \geq 0.5)$. Acc@0.5 measures overall correctness at this fixed threshold,

$$\mathrm{Acc@0.5} = \frac{\mathrm{TP}_{0.5} + \mathrm{TN}_{0.5}}{\mathrm{TP}_{0.5} + \mathrm{TN}_{0.5} + \mathrm{FP}_{0.5} + \mathrm{FN}_{0.5}}, \tag{36}$$

where the subscript 0.5 indicates counts computed after thresholding at 0.5. F1@0.5 is the harmonic mean of precision and recall at the same fixed threshold,

$$\mathrm{F1@0.5} = \frac{2\,\mathrm{Precision}_{0.5}\,\mathrm{Recall}_{0.5}}{\mathrm{Precision}_{0.5} + \mathrm{Recall}_{0.5}} = \frac{2\,\mathrm{TP}_{0.5}}{2\,\mathrm{TP}_{0.5} + \mathrm{FP}_{0.5} + \mathrm{FN}_{0.5}}. \tag{37}$$

Thresholded accuracy and F1 must always be interpreted together with their threshold. If a result uses a validation-selected threshold τ_{val} instead of 0.5, we denote it as Acc(val-thr) or F1(val-thr). In that setting, τ_{val} is selected only on the validation split and then applied unchanged to the test split. For TAMR-DTI, repeated-run summaries are reported as mean $\pm$ standard deviation across five seeds. We also report 95% confidence intervals computed as $\bar{x} \pm t_{0.975,4} s/\sqrt{5}$, with $t_{0.975,4} = 2.776$. Because the reported baselines do not provide per-seed predictions, we do not compute p-values against those baselines.

4.1.3 Experimental Configuration

The architecture of TAMR-DTI is described in Section 3; this subsection reports only the hardware, software, training, optimization, and evaluation settings used in the main repeated-seed experiments. Table 2 summarizes the compute environment recorded in the main experiment logs. All datasets use the same optimizer, scheduler, checkpoint-selection rule, and validation-only threshold-selection protocol for Acc(val-thr) and F1(val-thr). The classification threshold is searched on the validation set over a fixed grid from 0.01 to 0.99 and then applied unchanged to the test set.

The weighted binary cross-entropy loss is applied to the logit z_i for each drug-target pair:

$$\mathcal{L}_{\mathrm{BCE}} = -\frac{1}{N} \sum_{i=1}^{N} \left[w_+ y_i \log \sigma(z_i) + (1 - y_i) \log\left(1 - \sigma(z_i)\right) \right], \tag{38}$$

where $y_i \in \{0, 1\}$ is the interaction label, $\sigma(\cdot)$ is the sigmoid function, and w_+ is the positive-class weight. When automatic positive weighting is enabled, the positive-class weight is computed separately for each dataset as

$$w_+ = \frac{N_-^{\mathrm{train}}}{N_+^{\mathrm{train}}}, \tag{39}$$

Table 2: Hardware and software environment for the main repeated-seed experiments. The values are taken from the recorded run metadata and package snapshots.

Item	Value
CPU	Intel Xeon E5-2686 v4 @ 2.30GHz; 72 logical CPU cores
System memory	252 GB RAM
GPU	8 $\times$ NVIDIA Tesla P100-PCIE-16GB; 16 GB memory per GPU
GPU driver and CUDA	NVIDIA driver 535.288.01; CUDA 12.1 reported by the runtime metadata; PyTorch build uses CUDA 12.4 packages
Operating system	Linux 5.15.0-174-generic x86_64 with glibc 2.35
Python	Python 3.9.25 in the `ldm-dti` conda environment
Core deep-learning stack	PyTorch 2.4.0+cu124, DeepSpeed 0.18.9, DGL 2.4.0+cu124, and DGLLife 0.3.2
Chemistry and ML libraries	RDKit 2025.9.2, scikit-learn 1.6.1, NumPy 1.26.4, pandas 2.3.3, and SciPy 1.13.1
Sequence/modeling dependencies	transformers 4.57.6, mamba-ssm 2.2.2, and causal-conv1d 1.6.1
Experiment tracking	SwanLab 0.7.16

Table 3: Training and evaluation hyperparameters for the main experiments.

Hyperparameter	Value
Data split	Fixed random train/validation/test split; split sizes and label ratios are reported in Appendix Table 9
Optimizer	Adam with PyTorch default momentum parameters ($\beta_1 = 0.9, \beta_2 = 0.999, \epsilon = 10^{-8}$)
Training objective	Weighted binary cross-entropy with logits; positive-class weight computed from the training split
Initial learning rate	1×10^{-4}
Weight decay	1×10^{-4}
Maximum epochs	100
Micro-batch size	16 per GPU
Gradient accumulation	4 steps
Distributed training	DeepSpeed ZeRO-2; 8 GPUs in the main launch configuration
DeepSpeed details	Contiguous gradients, communication overlap, reduce-scatter, and all-gather partitioning enabled; all-gather and reduce bucket sizes set to 2×10^8
Effective batch size	512 drug-target pairs when using 8 GPUs
Gradient clipping	1.0
Mixed precision	Disabled in the reported runs
Auxiliary loss weights	Field-entropy loss $= 0.0$
Data-loader workers	8 per process
Learning-rate scheduler	Plateau scheduler
Learning-rate decay factor	0.5
Scheduler patience	5 validation epochs without sufficient improvement
Minimum learning rate	1×10^{-6}
Checkpoint-selection metric	Validation AUROC
Minimum improvement for checkpoint update	1×10^{-4}
Early-stopping patience	10 validation epochs without sufficient improvement
Threshold search grid	$\{0.01, 0.02, \ldots, 0.99\}$ on the validation split
Threshold-selection metric	Validation Acc(val-thr) for the main AUROC-selected runs
Reported seeds	42, 43, 44, 45, and 47
Repeated-run summary	Mean $\pm$ standard deviation over the five reported seeds
95% confidence interval	$\bar{x} \pm 2.776\, s/\sqrt{5}$, using the t critical value with four degrees of freedom

where N_+^{train} and N_-^{train} are the numbers of positive and negative samples in the training split. Table 4 reports the resulting values. This setting reduces the effect of class imbalance on BindingDB and Human while leaving the near-balanced BioSNAP and C. elegans datasets almost unchanged.

Table 4: Dataset-specific positive-class weights used by the binary cross-entropy loss.

Dataset	Training positives	Training negatives	Positive-class weight
BindingDB	14,583	19,856	1.3616
BioSNAP	9,684	9,540	0.9851
Human	1,846	2,351	1.2736
C. elegans	2,727	2,723	0.9985

4.2 Baseline Methods

SVM (Cortes & Vapnik, 1995) is included as a reported classical machine-learning baseline. We adopt its values, together with all other baseline values in this section, from the reported comparison of Ji et al. (2026) rather than reimplementing or retraining the baseline methods. SVM provides a maximum-margin classifier for separating drug-target feature vectors with a high-dimensional decision boundary. In the DTI setting, it serves as a non-neural reference for whether fixed pairwise features are already separable without learned molecular or protein encoders.

RF (Breiman, 2001) is included as a reported classical machine-learning baseline based on tree ensembles. It aggregates multiple decision trees trained on bootstrapped samples and feature subsets, with the reported comparison using 100 trees. This baseline can capture nonlinear feature interactions without gradient-based representation learning, making it a useful classical reference against graph and attention models. Its inclusion also helps separate gains from learned interaction modeling from gains that may come from strong tabular-style feature partitioning.

KNN (Guo et al., 2003) is included as a reported classical machine-learning baseline based on instance-level similarity. It predicts a query drug-target pair from nearby training examples, with the reported comparison using $k = 5$. Unlike parametric neural models, KNN relies directly on the geometry of the feature space and therefore reflects how much local similarity among drug-target pairs is sufficient for prediction. This makes it a simple but informative reference for datasets where similar drugs or targets may recur across splits.

LR (Friedman et al., 2000) is included as a reported classical machine-learning baseline based on linear probabilistic classification. It maps weighted input features to an interaction probability through a sigmoid function. Because LR has limited capacity, its reported performance provides a lower-complexity reference for the discriminative strength of the input features alone. It also gives a calibrated probabilistic baseline for comparison with higher-capacity neural interaction models.

GraphDTA (Nguyen et al., 2021) is included as a reported graph-based neural baseline. It encodes molecular graphs with graph neural networks and protein sequences with convolutional layers before fully connected interaction prediction. This design represents a common DTI modeling pattern in which ligand topology and protein sequence motifs are encoded separately and fused only after feature extraction. It is therefore a relevant reference for evaluating whether more explicit interaction modeling improves over standard graph-plus-sequence encoding.

DrugBAN (Bai et al., 2023) is included as a reported graph- and attention-based DTI baseline. It uses bilinear attention to match drug and target representations and incorporates domain adaptation to improve transfer across data distributions. The bilinear attention module gives an explicit mechanism for pairwise drug-target matching, rather than relying only on concatenated representations. Its domain-adaptation component also makes it a useful reference for the cross-dataset distribution differences considered in this study.

IIFDTI (Cheng et al., 2022) is included as a reported attention-based DTI baseline. It separates interactive and independent features before predicting drug-target interactions. This separation is intended to distinguish pair-specific interaction evidence from features that mainly describe each molecule or protein independently. We include it as a reported reference for attention-based feature disentanglement in DTI prediction.

BINDTI (Peng et al., 2025) is included as a reported structure-aware attention baseline. It combines GCN-based drug graph encoding, ACmix-based protein encoding, and bidirectional multi-head interaction attention. Its bidirectional attention design explicitly models how drug-side and target-side representations attend to each other during prediction. This makes it one of the closest reported references for comparing against models that emphasize interaction-aware fusion.

TransformerCPI (Chen et al., 2020) is included as a reported Transformer-based DTI baseline. It applies sequence self-attention to compound-protein interaction prediction and evaluates the learned interaction patterns with label-reversal experiments. The model emphasizes attention over sequential representations, providing a contrast to graph-centered molecular encoders. It is included to cover DTI baselines where long-range dependency modeling is handled primarily through Transformer-style attention.

MolTrans (Huang et al., 2021) is included as a reported Transformer-based interaction baseline. It tokenizes drug and protein substructures and models their higher-order interactions with Transformer and convolutional modules. By operating on substructure-level tokens, MolTrans provides a different interaction granularity from whole-graph or whole-sequence encoders. This makes it a useful reported reference for evaluating whether explicit substructure interaction modeling is competitive with the proposed target-aware refinement strategy.

LDM-DTI (Ji et al., 2026) is included as the reported multimodal reference model. It combines pretrained drug and protein representations with graph topology, molecular geometry, protein local-context encoding, and bidirectional interaction fusion. We use it as a close reported reference for multimodal DTI encoding and interaction modeling, while avoiding any claim that it was reproduced in our experiments. Table 5 therefore serves as a reported ranking-metric landscape rather than a paired significance comparison: the baseline rows are literature values, whereas TAMR-DTI is evaluated under our repeated-run protocol.

Table 5: AUROC/AUPRC comparison on four benchmark datasets. Baseline values are adopted from reported results; TAMR-DTI values are from our five-seed runs and are shown as mean $\pm$ standard deviation.

	BindingDB		BioSNAP		Human		C. elegans	
Method	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC
SVM	0.904 ± 0.002	0.865 ± 0.001	0.819 ± 0.045	0.839 ± 0.038	0.913 ± 0.012	0.905 ± 0.001	0.925 ± 0.009	0.907 ± 0.002
RF	0.942 ± 0.001	0.923 ± 0.001	0.857 ± 0.009	0.872 ± 0.006	0.939 ± 0.009	0.927 ± 0.001	0.935 ± 0.009	0.931 ± 0.001
KNN	0.895 ± 0.001	0.865 ± 0.002	0.842 ± 0.003	0.805 ± 0.003	0.915 ± 0.002	0.902 ± 0.002	0.912 ± 0.003	0.896 ± 0.002
LR	0.916 ± 0.003	0.884 ± 0.002	0.821 ± 0.004	0.796 ± 0.002	0.895 ± 0.003	0.862 ± 0.001	0.892 ± 0.002	0.883 ± 0.003
GraphDTA	0.944 ± 0.004	0.925 ± 0.006	0.871 ± 0.004	0.870 ± 0.006	0.965 ± 0.003	0.955 ± 0.003	0.959 ± 0.004	0.951 ± 0.002
DrugBAN	0.952 ± 0.001	0.936 ± 0.001	0.902 ± 0.001	0.905 ± 0.002	0.979 ± 0.001	0.969 ± 0.005	0.981 ± 0.002	0.978 ± 0.002
IIFDTI	0.924 ± 0.002	0.901 ± 0.004	0.895 ± 0.003	0.898 ± 0.005	0.977 ± 0.003	0.967 ± 0.027	0.984 ± 0.002	0.979 ± 0.002
TransformerCPI	0.895 ± 0.002	0.861 ± 0.003	0.843 ± 0.008	0.859 ± 0.007	0.956 ± 0.005	0.943 ± 0.002	0.975 ± 0.003	0.975 ± 0.002
MolTrans	0.935 ± 0.003	0.901 ± 0.004	0.879 ± 0.006	0.882 ± 0.006	0.979 ± 0.003	0.975 ± 0.002	0.982 ± 0.002	0.979 ± 0.002
BINDTI	0.956 ± 0.001	0.942 ± 0.002	0.895 ± 0.003	0.894 ± 0.003	0.980 ± 0.001	0.975 ± 0.004	0.985 ± 0.002	0.984 ± 0.002
LDM-DTI	0.960 ± 0.002	0.945 ± 0.002	0.908 ± 0.003	0.906 ± 0.003	0.981 ± 0.001	0.976 ± 0.002	0.988 ± 0.002	0.986 ± 0.002
TAMR-DTI	0.964 ± 0.001	0.954 ± 0.002	0.930 ± 0.002	0.935 ± 0.002	0.982 ± 0.002	0.978 ± 0.002	0.992 ± 0.001	0.991 ± 0.001

4.3 Benchmark Performance and Dataset-Level Behavior

The benchmark results support a qualified, dataset-dependent conclusion. TAMR-DTI improves threshold-independent ranking performance on BindingDB, BioSNAP, Human, and C. elegans, with the clearest margin on BioSNAP and the smallest margin on Human. The Human benchmark therefore changes the interpretation from ranking underperformance to a calibration and stability limitation: its AUROC/AUPRC are at or above the reported-reference level, but its fixed-threshold scores remain lower. Because the baseline values are adopted from prior reports rather than reproduced under a paired protocol, we use them as reference points for positioning the model, not as the basis for significance testing.

The AUROC/AUPRC landscape in Table 5 is the main comparison in this section. TAMR-DTI reaches the strongest reported ranking scores on the four benchmarks: BindingDB improves from the reported 0.960/0.945 AUROC/AUPRC envelope to 0.964/0.954, BioSNAP improves more substantially to 0.930/0.935, Human increases to 0.982/0.978, and C. elegans increases to 0.992/0.991. These gains are most meaningful for ranking drug-target pairs, where the decision threshold is not fixed in advance and the model is judged by how well it separates interacting from non-interacting pairs across operating points.

BioSNAP provides the central positive case for TAMR-DTI. It is nearly balanced, has enough repeated drug-target structure for multimodal interaction modeling to be useful, and shows consistent improvements across AUROC, AUPRC, Acc@0.5, and F1@0.5. BindingDB and C. elegans also support the ranking-metric claim, but their fixed-threshold behavior is more mixed: BindingDB improves in AUROC, AUPRC, and Acc@0.5 while F1@0.5 decreases, and C. elegans mainly shows ranking gains with only small changes in thresholded metrics. We therefore avoid describing the model as uniformly superior across every metric and instead emphasize where the evidence is strongest.

Human behaves differently. The dataset is the smallest benchmark in this study and has a more difficult entity-support profile, so repeated seeds show larger variation, especially for fixed-threshold metrics. On the Human random split, TAMR-DTI reaches 0.982 ± 0.002 AUROC and 0.978 ± 0.002 AUPRC, exceeding the strongest reported ranking references at the displayed precision. However, its 0.930 ± 0.009 Acc@0.5 and 0.924 ± 0.009 F1@0.5 remain below the strongest reported thresholded scores. The Human result is therefore useful in a more specific way: it shows that the architecture can rank Human interactions competitively, while calibration and fixed-threshold robustness still need improvement on small, sparse Human splits.

Table 6 reports the same TAMR-DTI repeated-run results across all four metrics. This table is included to make the fixed-threshold behavior explicit rather than relying only on the visual summary.

Table 6: TAMR-DTI repeated-run summary across the four benchmark datasets. Values are mean $\pm$ standard deviation over seeds 42, 43, 44, 45, and 47.

Dataset	AUROC	AUPRC	Acc@0.5	F1@0.5
BindingDB	0.964 ± 0.001	0.954 ± 0.002	0.909 ± 0.003	0.891 ± 0.004
BioSNAP	0.930 ± 0.002	0.935 ± 0.002	0.857 ± 0.004	0.859 ± 0.004
Human	0.982 ± 0.002	0.978 ± 0.002	0.930 ± 0.009	0.924 ± 0.009
C. elegans	0.992 ± 0.001	0.991 ± 0.001	0.961 ± 0.004	0.961 ± 0.004

The same pattern is summarized visually in Figure 2. The figure includes all reported baselines as reference points and places TAMR-DTI as a five-seed mean with one-standard-deviation error bars. It is intended to show model positioning across datasets and metrics, not to imply a paired statistical test against any single prior method.

Figure 3 provides a complementary uncertainty view for AUPRC, Acc@0.5, and F1@0.5. For the reported baselines, intervals are computed from the published mean and standard deviation using the same five-run t-interval formula when those summaries are available; for TAMR-DTI, intervals are computed from our five seeds. This plot is again descriptive rather than paired: it shows whether the TAMR-DTI estimates sit outside or near the reported uncertainty ranges, while preserving the distinction between literature baselines and our own repeated runs.

The fixed-threshold results are reported in Table 6 and Figure 2 for completeness, but they are secondary to the main ranking-metric claim. Acc@0.5 and F1@0.5 depend on score calibration and on the fixed threshold used for all methods, so they are more sensitive to class balance and split composition. This explains why BindingDB and C. elegans show clearer improvements in AUROC/AUPRC than in every thresholded metric, and why Human improves the ranking references while still lagging behind the strongest reported fixed-threshold scores.

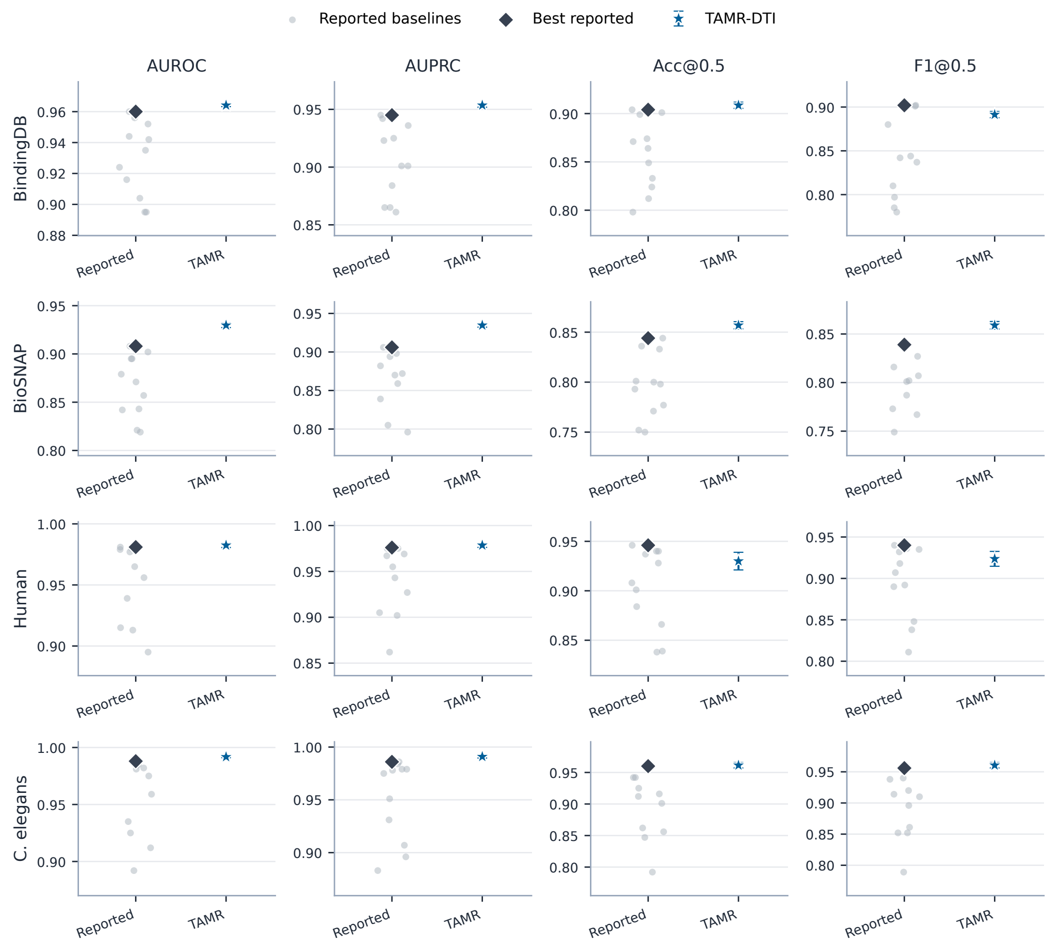

Figure 2: Main result comparison across four datasets and four metrics. Gray points denote reported baselines, dark diamonds denote the best reported value, and blue stars denote TAMR-DTI five-seed means with one-standard-deviation error bars. Acc@0.5 and F1@0.5 are fixed-threshold metrics.

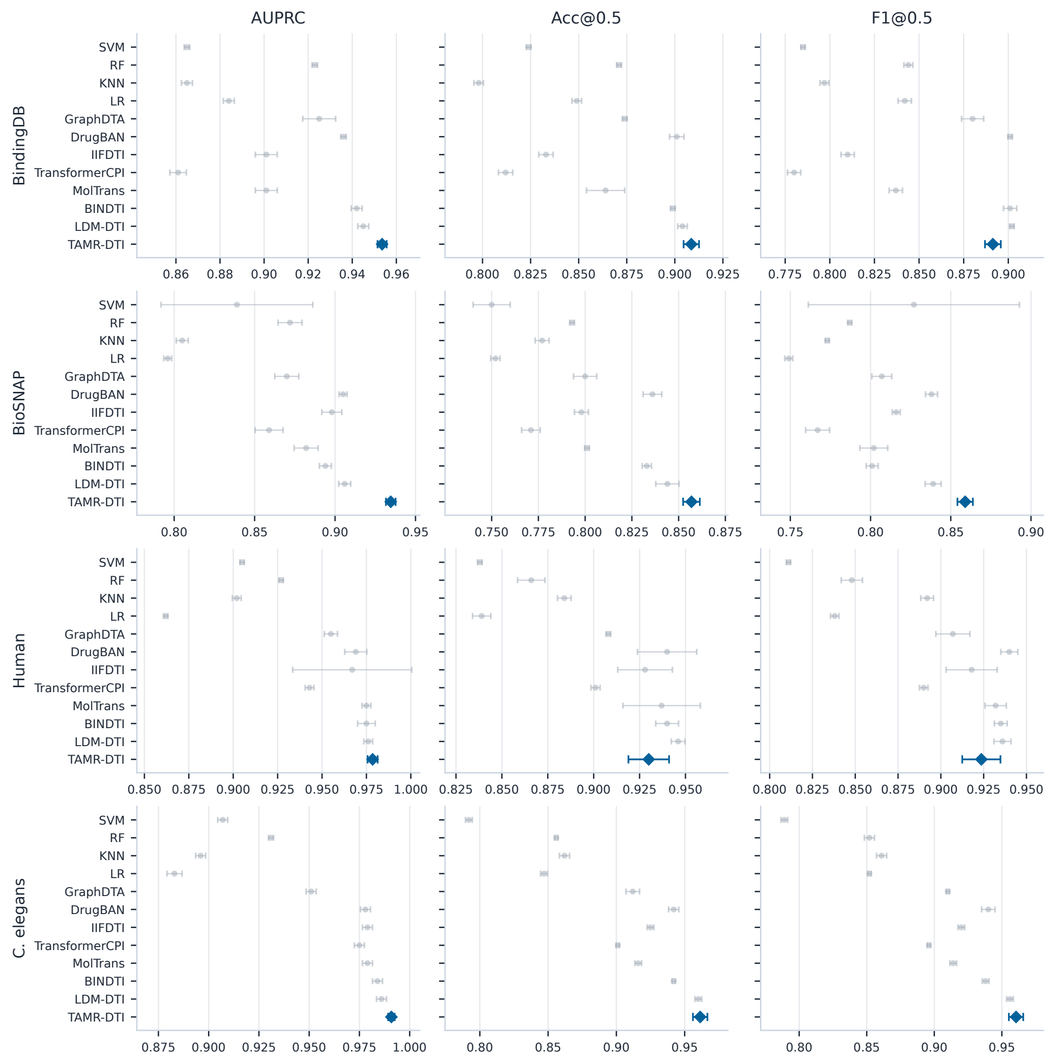

Figure 3: All-method uncertainty comparison on AUPRC, Acc@0.5, and F1@0.5. Baseline intervals are derived from reported mean and standard deviation values, whereas TAMR-DTI intervals are computed from five local seeds.

4.4 Stability Across Seeds

The repeated-seed analysis separates stable ranking gains from split-sensitive behavior. BindingDB, BioSNAP, and C. elegans form compact AUROC/AUPRC clusters, indicating that the ranking improvements are not driven by a single favorable seed. Human also stays above the reported-reference ranking range across most seeds, but its fixed-threshold metrics vary more strongly than the larger benchmarks. This is why the Human result is discussed as a calibration and stability limitation even though its mean AUROC and AUPRC are competitive with the strongest reported references.

The seed-level view in Figure 4 is used for this interpretation. The plot is descriptive rather than inferential: it shows where the proposed model is stable enough to support a benchmark claim and where additional calibration and repeated-run robustness work is still needed.

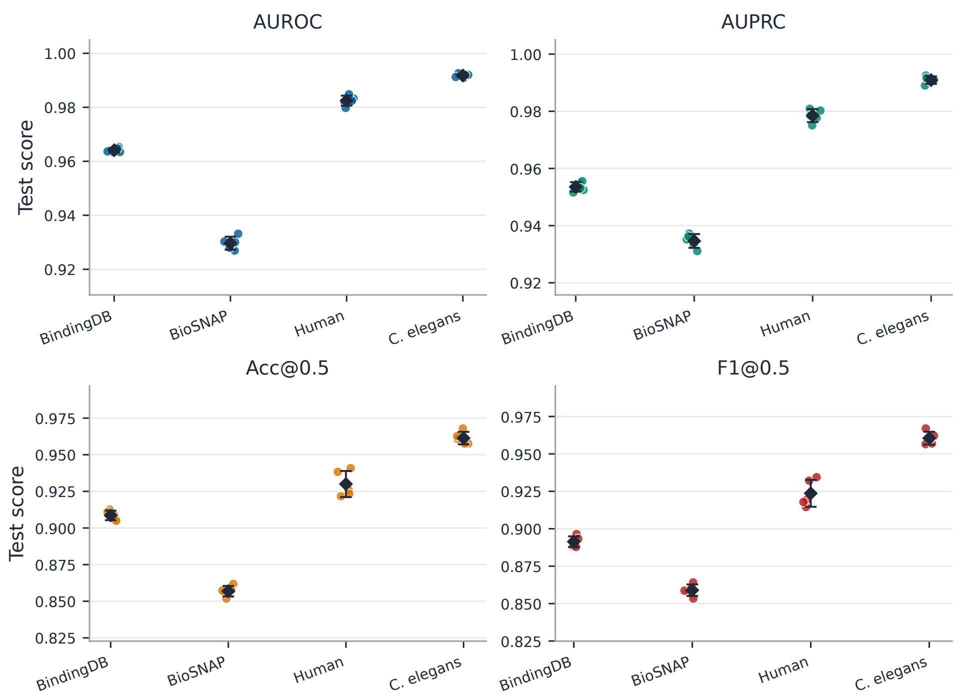

Figure 4: Five-seed stability of TAMR-DTI on the four benchmark datasets. Dots denote individual seeds 42, 43, 44, 45, and 47; diamonds denote seed means; and error bars denote one standard deviation. Acc@0.5 and F1@0.5 use the fixed threshold 0.5.

4.5 Ablation, Conformer Comparison, and Interpretability

The BioSNAP gain is analyzed through the mechanism that TAMR-DTI is designed to implement rather than through separate table-by-table comparisons. The initial protein representation first conditions the selection of drug conformers, so the small-molecule geometry branch is not averaged independently of the target. The aggregated drug context then feeds back into protein encoding through FiLM modulation, protein-side Mamba refines the longer sequence-token stream, and the final bidirectional interaction module aligns the two modalities before prediction. We test this chain from three internal perspectives: module removal, conformer-count control, and model-internal evidence from conformer weights, FiLM strength, and Mamba refinement. BioSNAP is used because it is nearly balanced and gives the clearest ranking-metric improvement in the main benchmark comparison.

4.5.1 Module ablation

The module ablation asks whether the BioSNAP result comes from the intended cross-modal conditioning path or from a single oversized encoder. We conduct a single-seed diagnostic study with seed

42, the same train/validation/test split, the same validation-AUROC checkpoint-selection rule, and the same training configuration as the full model. Each ablated variant removes one module while keeping the remaining architecture unchanged. Because the ablation is not repeated over multiple seeds, we treat it as a diagnostic module-necessity analysis and report absolute scores rather than inferential claims. Acc@0.5 and F1@0.5 are computed at the fixed threshold 0.5.

The full model forms the upper envelope in Table 7, reaching 0.933 AUROC, 0.937 AUPRC, 0.862 Acc@0.5, and 0.864 F1@0.5. The largest AUROC loss appears when target-aware conformer aggregation is removed, dropping from 0.933 to 0.916 and reducing F1@0.5 from 0.864 to 0.845. This supports the core geometry argument: conformers should not be treated as target-agnostic views, because the relevant molecular geometry depends on the protein context. Removing ligand-conditioned protein FiLM gives the largest fixed-threshold accuracy drop, from 0.862 to 0.842, which is consistent with the feedback direction in the architecture: after protein-guided conformer aggregation, the resulting drug context should still be allowed to modulate protein encoding before final interaction prediction.

Table 7: Single-seed diagnostic ablation results on BioSNAP. All variants use seed 42, the same split, validation-AUROC checkpoint selection, and fixed-threshold Acc@0.5/F1@0.5.

Variant	AUROC	AUPRC	Acc@0.5	F1@0.5
Full TAMR-DTI	**0.933**	**0.937**	**0.862**	**0.864**
w/o Target-aware conformer	0.916	0.922	0.844	0.845
w/o Protein FiLM	0.920	0.924	0.842	0.844
w/o Protein Mamba refinement	0.918	0.925	0.842	0.845
w/o Bidirectional modulation	0.920	0.926	0.848	0.849

Reading the ablation scores as drops from the full model makes the component roles clearer. The relative-drop view in Figure 5 shows that removing the target-aware conformer module produces the largest AUROC drop, removing ligand-conditioned FiLM produces the largest Acc@0.5 drop, and removing protein Mamba refinement or bidirectional modulation yields consistent but slightly smaller degradation across the four metrics. The full model is included as the zero-drop reference, so the plot directly shows how far each ablated configuration moves from the complete TAMR-DTI setting.

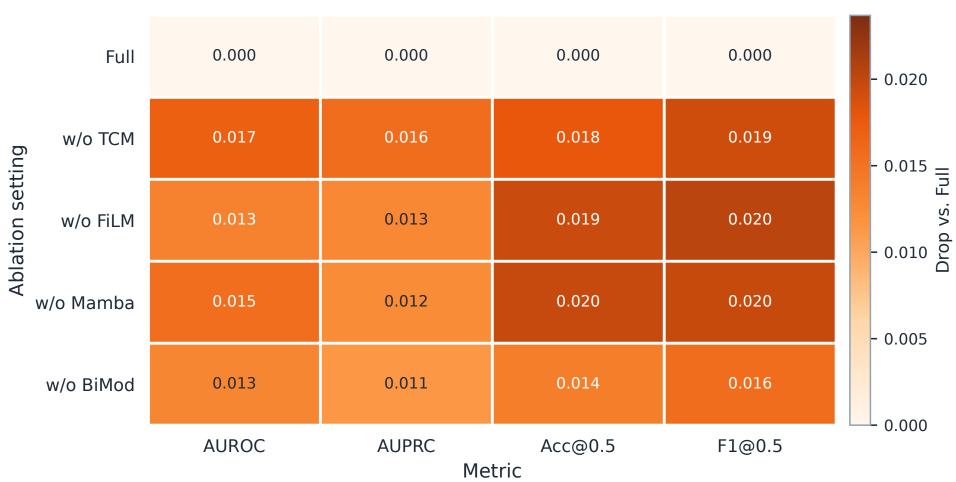

Figure 5: Single-seed diagnostic ablation drops on BioSNAP relative to the full TAMR-DTI model. Each cell reports the relative test-score decrease caused by removing one module under the same seed, split, validation-AUROC checkpoint selection, and fixed-threshold Acc@0.5/F1@0.5; the full model is shown as the zero-drop reference.

The protein Mamba and bidirectional-interaction ablations complete the same picture. Removing protein Mamba refinement lowers AUROC to 0.918 and F1@0.5 to 0.845, showing that the protein branch loses discriminative information when its token sequence is not further refined. This placement is intentional: TAMR-DTI applies Mamba to protein tokens rather than to SMILES tokens, because the drug branch is already represented through graph, pretrained, and conformer-level views, while the protein branch provides the longer sequence-token substrate where state-space refinement has a clearer role. Removing bidirectional modulation gives the smallest but still consistent degradation, with AUROC decreasing to 0.920 and F1@0.5 decreasing to 0.849. The four modules therefore act as complementary parts of one conditioning path: target-aware geometry selection, drug-to-protein FiLM feedback, protein-side sequential refinement, and final cross-modal alignment.

4.5.2 Conformer-count comparison

The conformer-count comparison isolates where the geometry branch helps. We compare six BioSNAP seed-42 variants that differ in the effective number of valid conformers or in the conformer aggregation rule. The $n = 1$, $n = 2$, and $n = 4$ variants are derived from the same eight-slot conformer cache by keeping only the first n valid conformer slots and masking the rest. The $n = 8$ variant uses the full eight-conformer cache used in the main model, and the $n = 16$ variant uses a separately generated sixteen-conformer cache to test whether a larger candidate geometry set improves the model further. Because preprocessing sorts generated conformers by computed energy, conformer count also controls the energy range exposed to the model: $n = 1$ keeps only the lowest-energy conformer, $n = 2$ and $n = 4$ add lower-energy alternatives, $n = 8$ exposes a broader geometry set, and $n = 16$ adds still more candidate conformers that may also contain additional irrelevant geometry. The $n = 8$ uniform-average variant keeps the same eight-conformer input but disables target-aware scoring and therefore serves as the aggregation control. All variants use the same data split, seed, optimizer, and validation-AUROC checkpoint selection; this experiment is interpreted as a conformer-count diagnostic rather than a hyperparameter search. The early-stopping patience is set to 100 epochs in this diagnostic comparison so that all variants can complete the full training horizon; Acc@0.5 and F1@0.5 are computed at the fixed threshold 0.5.

Table 8: Single-seed conformer-count comparison on BioSNAP. Conformers are energy-sorted during preprocessing; $n = 1$ keeps the lowest-energy conformer only, while larger n values expose additional geometries.

Variant	Effective conformers	Aggregation	Best epoch	AUROC	AUPRC	Acc@0.5	F1@0.5
$n = 1$ target-aware	1	Target-aware	46	0.9277	0.9297	0.8578	0.8591
$n = 2$ target-aware	2	Target-aware	68	0.9300	0.9322	0.8575	0.8605
$n = 4$ target-aware	4	Target-aware	46	0.9297	0.9345	0.8524	0.8542
$n = 8$ target-aware	8	Target-aware	51	**0.9322**	**0.9356**	0.8575	0.8624
$n = 16$ target-aware	16	Target-aware	62	0.9312	0.9326	**0.8631**	**0.8668**
$n = 8$ uniform average	8	Uniform average	64	0.9308	0.9340	0.8589	0.8616

The results in Table 8 show that the eight-conformer target-aware variant gives the strongest ranking performance, reaching 0.9322 AUROC and 0.9356 AUPRC. The single-conformer variant reaches 0.9277 AUROC and 0.9297 AUPRC, which suggests that fixing the drug geometry to the lowest-energy conformer discards useful alternative spatial views. Increasing the candidate set to $n = 16$ does not improve the ranking metrics over $n = 8$: AUROC decreases from 0.9322 to 0.9312 and AUPRC decreases from 0.9356 to 0.9326, although fixed-threshold Acc@0.5 and F1@0.5 increase in this single-seed run. This explains why the main experiments use $K = 8$: it provides the best AUROC/AUPRC tradeoff for the ranking-focused benchmark claim, while avoiding the additional geometry noise and compute cost introduced by sixteen conformers. The intermediate $n = 2$ and $n = 4$ variants also do not improve monotonically across all metrics, so the result should not be read as ”more conformers always win.” Instead, it supports a more specific interpretation: energy-sorted conformers provide a candidate geometry set, and target-aware aggregation is needed to decide which conformers are useful for a given protein.

The absolute-score view in Figure 6 highlights the same tradeoff. The target-aware $n = 8$ variant is highest on AUROC and AUPRC, whereas $n = 16$ is higher on the fixed-threshold Acc@0.5 and F1@0.5 in this diagnostic run. We therefore interpret the conformer-count comparison conserva-

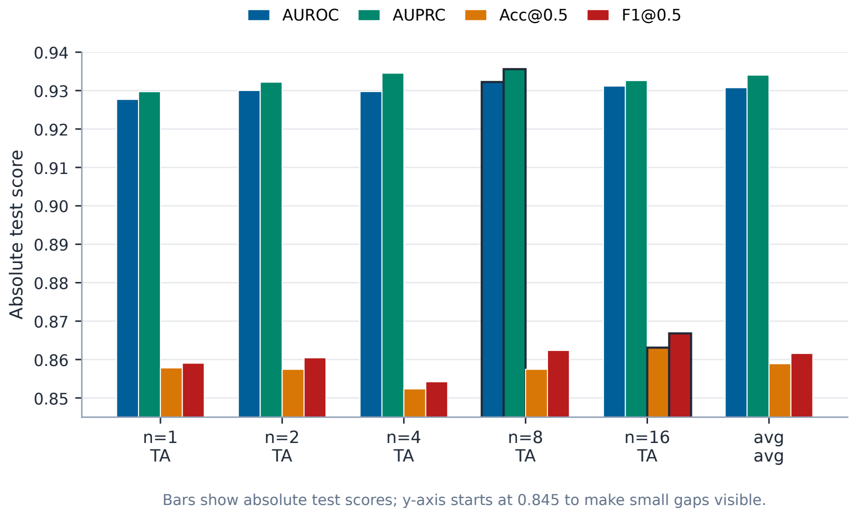

Figure 6: Absolute-score comparison of BioSNAP conformer-count variants, including the sixteen-conformer diagnostic run. TA denotes target-aware conformer aggregation, and avg denotes uniform averaging. The y-axis starts from 0.845 to make the small metric gaps visible; Acc@0.5 and F1@0.5 use the fixed threshold 0.5.

tively: multiple conformers are useful, and protein-conditioned weighting improves ranking metrics up to the eight-conformer setting, but the fixed-threshold metrics do not support a claim that a larger conformer set dominates at every operating point.

Together with the benchmark comparison, these diagnostics explain why BioSNAP is the strongest positive case for TAMR-DTI. The gain is not attributed to a single module in isolation. It depends on protein-guided conformer selection, drug-context feedback into protein encoding, protein-side sequential refinement, and final bidirectional alignment. The conformer-count result also places the energy issue in the right scope: the lowest-energy conformer is a useful starting point, but binding prediction benefits from exposing additional geometries and letting the target representation weight them.

4.5.3 Interpretability analysis

The ablation study shows that the proposed modules are useful, but it does not directly show how the full model uses them during inference. We therefore conduct a module-oriented interpretability analysis on the BioSNAP seed-42 model, using the same best checkpoint and the validation-selected threshold of 0.60 only to assign prediction groups in the diagnostic plots. The analysis covers all 5,493 BioSNAP test pairs and focuses on three quantities that are directly tied to the proposed architecture: target-aware conformer weights, ligand-conditioned FiLM modulation strength, and protein Mamba refinement magnitude. Since the protein features are pooled token representations rather than experimentally annotated residue coordinates, we interpret the visualizations as model-internal evidence instead of claiming exact biochemical binding sites.

For the target-aware conformer module, the model produces a distribution over the K valid conformers of a drug molecule. For a test pair i, let $\mathbf{w}_i = \{w_{i1}, \ldots, w_{iK}\}$ denote the conformer weights. We quantify the sharpness of conformer selection by the normalized entropy

$$\bar{H}_i = -\frac{1}{\log K_i} \sum_{k=1}^{K_i} w_{ik} \log(w_{ik} + \epsilon), \tag{40}$$

where K_i is the number of valid conformers and ϵ is a small constant for numerical stability. A smaller $\bar{H}_i$ indicates that the prediction depends more strongly on a small subset of conformers, whereas a larger value indicates softer multi-conformer aggregation.

Representative conformer-weight maps in Figure 7 show that the model does not use a single fixed conformer pattern for all inputs. Some pairs concentrate almost all mass on one conformer, while others distribute probability over two, three, or more valid conformers. Over the full test set, the mean top-1 conformer weight is 0.241 and the mean top-1 margin is 0.069. The mean normalized conformer entropy is 0.931, indicating that the model often keeps multiple conformers active rather than forcing a hard one-hot choice. Error cases are more concentrated on average: false positives and false negatives have lower normalized entropy, 0.898 and 0.879, and higher top-1 weights, 0.267 and 0.284, respectively. This suggests that over-confident conformer selection can be one failure pattern, while correct predictions more often benefit from a softer target-aware conformer mixture.

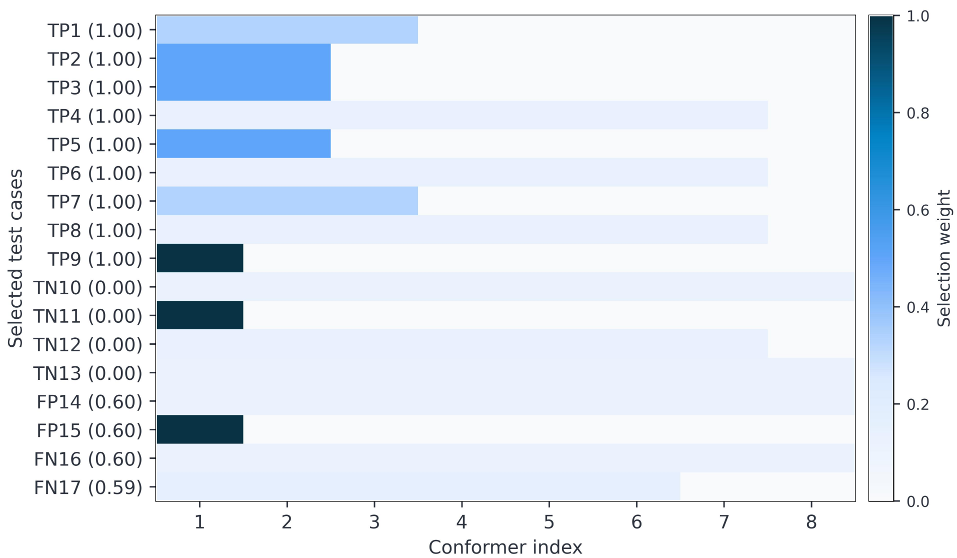

Figure 7: Target-aware conformer-weight visualization on selected BioSNAP test cases. Rows denote representative test pairs and columns denote the eight conformer slots. Color intensity indicates the conformer selection weight assigned by TAMR-DTI.

We next inspect the ligand-conditioned protein modulation module. In TAMR-DTI, the drug context controls FiLM parameters in the protein encoder, so the protein representation is updated as

$$\widetilde{\mathbf{H}}_p^{(l)} = \mathbf{H}_p^{(l)} \odot (1 + \boldsymbol{\gamma}_d^{(l)}) + \boldsymbol{\beta}_d^{(l)}, \tag{41}$$

where $\boldsymbol{\gamma}_d^{(l)}$ and $\boldsymbol{\beta}_d^{(l)}$ are generated from the drug-side context at protein layer l. We measure the layer-wise modulation strength as

$$M_l = \|\boldsymbol{\gamma}_d^{(l)}\|_2 + \|\boldsymbol{\beta}_d^{(l)}\|_2. \tag{42}$$

The FiLM statistics in the right panel of Figure 8 show that the drug-to-protein feedback signal is not negligible. The first protein encoding layer has the strongest average modulation, with $M_1 = 0.0263$, followed by $M_2 = 0.0160$ and $M_3 = 0.0166$. This pattern is consistent with the design goal of TAMR-DTI: drug-side information is injected early enough to condition protein feature extraction, and subsequent layers further transform the conditioned representation instead of treating the protein as a fixed static embedding.

For the protein Mamba refinement module, we compare the protein tokens before and after the Mamba block. Let $\mathbf{h}_t$ be the input token at protein position t, $\mathbf{h}_t^M$ be the corresponding Mamba-refined token, and $g = \sigma(\alpha)$ be the learned residual gate. We compute the gated refinement magnitude

$$R_t = g\|\mathbf{h}_t^M - \mathbf{h}_t\|_2. \tag{43}$$

The learned gate is 0.129 in the BioSNAP checkpoint, which means that the model does not overwrite the protein encoder representation, but adds a controlled Mamba-based residual. Across the full test set, the mean gated refinement magnitude is 0.121 and the mean maximum token-level refinement is 0.124. The middle panel of Figure 8 compares this statistic across true positives, true negatives, false positives, and false negatives; the values are close but not identical across groups, indicating that the Mamba branch acts as a stable sequence-refinement component rather than an unstable classifier shortcut.

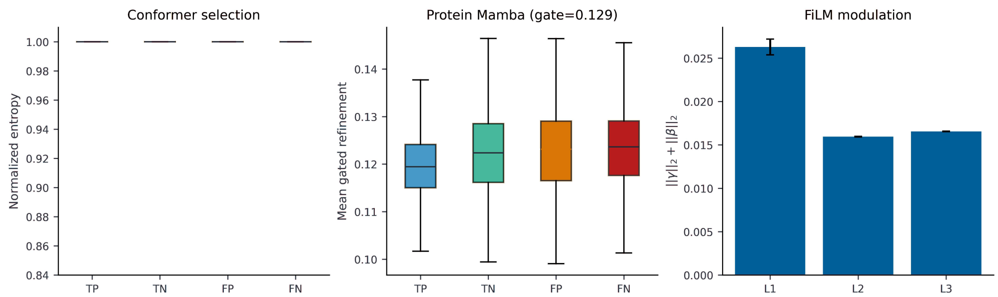

Figure 8: Module-level interpretability statistics on the BioSNAP test set. The conformer panel reports normalized conformer-weight entropy across prediction groups; the protein Mamba panel reports gated token-refinement magnitude; and the FiLM panel reports layer-wise ligand-conditioned modulation strength.

The token-level heatmap in Figure 9 further visualizes R_t along the 128 protein token positions for selected representative cases. The refinement is not uniformly distributed across all protein tokens. Instead, different drug-target pairs show different local peaks, which indicates that the Mamba block adjusts the protein representation in a sequence-position-dependent manner. This complements the ablation result in Table 7: removing protein Mamba refinement reduces BioSNAP performance, and the interpretability analysis shows that the retained module contributes a controlled, non-uniform refinement signal during inference.

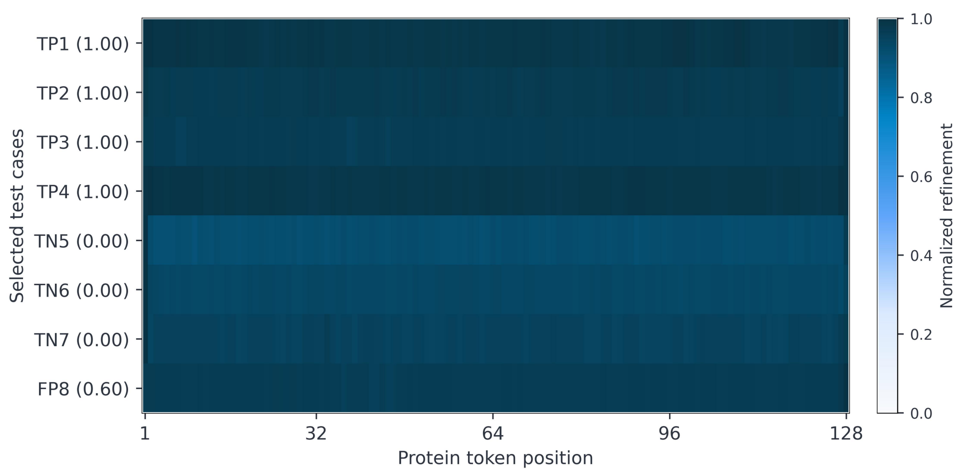

Figure 9: Protein Mamba refinement heatmap for selected BioSNAP test cases. Each row corresponds to a representative drug-target pair, the horizontal axis denotes pooled protein token positions, and color intensity denotes normalized gated refinement magnitude.

Overall, the interpretability results support the same conclusion as the ablation analysis from a different angle. Target-aware conformer aggregation exposes sample-specific conformer usage; FiLM confirms that protein encoding is conditioned by the drug context; and Mamba refinement provides a gated, token-dependent update to the protein sequence representation. These observations make the BioSNAP improvement more interpretable: TAMR-DTI does not rely only on a larger multimodal encoder, but uses explicit mechanisms to coordinate drug geometry, ligand-conditioned protein encoding, and sequential protein refinement before final interaction prediction.

5 DISCUSSION

The current evidence supports a conservative interpretation of TAMR-DTI. In the five-seed main experiments, the model improves the reported ranking metrics on BindingDB, BioSNAP, Human, and C. elegans, with the clearest gain on BioSNAP. At the same time, the Human random-split runs still show weaker fixed-threshold performance and larger seed-level variation than the larger benchmarks. The results therefore support a ranking-metric claim but not a claim of uniform dominance across all metrics. A more defensible conclusion is that target-aware conformer aggregation, ligand-conditioned protein modulation, and protein-side state-space refinement provide dataset-dependent benefits when the split distribution rewards pair-specific representation learning, while thresholded decisions remain sensitive to calibration.

The conformer results clarify why a target-aware design is useful. A single drug molecule should not necessarily be represented by one fixed 3D summary for all proteins. In the generated cache, the first conformer slot corresponds to the lowest-energy generated geometry, slots two to four provide lower-energy alternatives, and later slots contain higher-energy alternatives that may be less stable in isolation. The $n = 1$ experiment therefore approximates a fixed lowest-energy geometry baseline, whereas the $n = 2$, $n = 4$, $n = 8$, and $n = 16$ variants allow the model to consider broader geometric hypotheses. The target-aware $n = 8$ variant obtains the best AUROC and AUPRC, suggesting that alternative conformers can carry useful ranking information. The $n = 16$ variant instead obtains the best Acc@0.5 and F1@0.5 in this single-seed diagnostic run. This split between ranking and fixed-threshold metrics is plausible because a larger conformer set can change score calibration and the operating point around the fixed threshold, even when the additional geometries do not improve the overall positive-negative ranking. It may also introduce extra conformer hypotheses that help some borderline decisions while adding noise to pairwise ordering. This pattern is important: adding conformers alone is not the main claim. The benefit comes from allowing protein context to modulate which geometric views contribute to the final drug representation, while thresholded metrics remain affected by calibration at the fixed 0.5 threshold.

The protein-side Mamba component should be interpreted in the same restrained way. TAMR-DTI does not replace attention with Mamba throughout the architecture. Attention remains in ProteinACmix and BiIntention because DTI prediction still needs explicit pairwise matching between drug and protein tokens. Instead, Mamba is used as a gated residual refinement on the protein stream before the final interaction module. This placement matches the structure of the input: protein residues form an ordered biological sequence, so a selective state-space mixer can refine long-range sequential dependencies without constructing another global attention layer over the full fused token set. In contrast, the drug-side representation concatenates SMILES semantics, molecular graph topology, and conformer geometry, whose token order is more heterogeneous. Applying a single sequence propagation rule to mixed drug tokens, or replacing BiIntention with a generic sequence mixer, would risk imposing an artificial order and weakening the explicit cross-modal matching bias. The ablation and interpretability results support this design choice: removing protein Mamba reduces BioSNAP performance, while the learned gate keeps the refinement controlled rather than overwriting the pretrained protein representation.

The module ablations and interpretability analyses also suggest that the proposed components are complementary rather than redundant. Removing the target-aware conformer module causes the largest AUROC drop, consistent with the need to avoid target-independent geometry pooling. Removing ligand-conditioned FiLM gives the largest Accuracy drop, indicating that thresholded decisions benefit from conditioning protein features on drug context before final fusion. Removing Mamba weakens ranking performance, consistent with the role of protein-sequence refinement. The interpretability results provide a matching internal view: conformer weights are sample-specific rather than fixed, FiLM signals are nonzero across protein layers, and Mamba refinement varies across protein token positions. Together, these observations support the central idea that DTI prediction benefits when drug geometry and protein representations are conditioned on each other before the final classifier.

Several limitations remain. First, the main comparison is based on repeated seeds over fixed splits, whereas some reported baselines use different experimental protocols. This limits the strength of direct claims against previous work, especially when per-seed baseline outputs are unavailable. Second, the ablation and conformer-count analyses are single-seed diagnostic experiments on BioSNAP, so they should be read as mechanism evidence rather than as statistical significance tests. Third, the conformers are generated by computational preprocessing and ranked by force-field energy; they are useful geometric hypotheses, but they are not experimentally validated binding poses. Fourth, the Human result indicates that strong ranking performance does not automatically translate into equally strong fixed-threshold decisions on small and sparse splits. Future work should expand matched-seed ablations, evaluate more challenging scaffold or protein-family splits, study calibration-aware threshold selection, and incorporate stronger structural supervision when experimentally resolved binding information is available.

6 Conclusion

We presented TAMR-DTI, a multimodal DTI framework that represents drug-target pairs through target-aware modulation and protein-side Mamba refinement. The model addresses two limitations of fixed multimodal encoding: a ligand should not be forced into the same conformer summary for every target, and long-range protein refinement should not remove the explicit interaction modeling needed for DTI prediction. TAMR-DTI therefore lets protein context reweight molecular conformers, lets ligand context modulate protein tokens, and applies a gated Mamba residual block only to the ordered protein sequence before BiIntention cross-modal matching.

Across five repeated runs on four benchmark datasets, TAMR-DTI improves AUROC and AUPRC on BindingDB, BioSNAP, Human, and C. elegans, with the strongest improvement on BioSNAP. The Human benchmark still prevents a claim of uniform superiority because its fixed-threshold scores remain weaker and more variable than its ranking metrics. The BioSNAP ablation, conformer-count comparison, and interpretability analyses nevertheless provide consistent mechanism-level evidence: target-aware conformer aggregation improves ranking quality, ligand-conditioned protein modulation affects thresholded decisions, and protein-side Mamba refinement contributes a controlled sequence-dependent update.

Overall, TAMR-DTI shows that pair-specific molecular geometry and protein-only sequence refinement can be integrated without discarding explicit drug-target matching. The resulting model is best viewed as a competitive, dataset-dependent extension of a strong multimodal DTI backbone. Future work should strengthen the evidence with matched multi-seed ablations, harder generalization splits, and more direct structural supervision for conformer selection and binding-site interpretation.

Acknowledgments

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A Additional Dataset Statistics

For the main experiments, each dataset is split into training, validation, and test sets under a fixed random split. The training set is used for optimization, the validation set is used for checkpoint and threshold selection, and the test set is used only for final reporting. Table 9 reports the label distribution of each split. The split-level distribution is reported in the appendix because the main text focuses on aggregate dataset differences, while these counts document the exact processed split files used by the experiments.

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Table 9: Train/validation/test split distribution for the main experiments. Counts are computed from the processed split files used in this study.

Dataset	Split	Samples	Positive	Negative	Positive rate
BindingDB	Train	34,439	14,583	19,856	42.34%
BindingDB	Validation	4,920	2,017	2,903	41.00%
BindingDB	Test	9,840	4,074	5,766	41.40%
BioSNAP	Train	19,224	9,684	9,540	50.37%
BioSNAP	Validation	2,747	1,398	1,349	50.89%
BioSNAP	Test	5,493	2,748	2,745	50.03%
Human	Train	4,197	1,846	2,351	43.98%
Human	Validation	600	251	349	41.83%
Human	Test	1,200	536	664	44.67%
C. elegans	Train	5,450	2,727	2,723	50.04%
C. elegans	Validation	778	405	373	52.06%
C. elegans	Test	1,558	761	797	48.84%

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